The Bottleneck: A Gatekeeper Trained on Its Own Accepted Work
Narrows a Field While Making Its Bias Harder to Detect

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Abstract

A curator that forms its taste from work it has already accepted is training on a record of its own decisions. Rejections leave no trace, the counterfactual corpus is never observed, and the loop has no channel through which it could learn it was wrong. We model this as an agent-based simulation: creators propose work into a feature space, a gatekeeper accepts a fixed fraction, and the accepted set becomes the gatekeeper's training corpus for the next round. Across four conditions ($n = 40$ runs, 200 rounds each), the self-referential gatekeeper narrows the published corpus significantly further than a fixed gatekeeper of identical strictness (mean pairwise distance 0.423 against 0.496, $p = 0.0001$), and both fall far below an open control (0.804). The unanticipated result concerns detection. Survivor bias, the correlation between a creator's distance from the gatekeeper's founding taste and that creator's acceptance rate, is -0.937 under the fixed gatekeeper and -0.497 under the adaptive one. The filter producing the narrower corpus is the filter whose exclusions are harder to see. A reserved random-acceptance quota recovers part of the lost breadth ($+0.100$, $p = 0.0001$) and leaves survivor bias unchanged. A parameter sweep locates a scope condition: where creators hold no commitments of their own, the extra narrowing disappears.

1 Introduction

Consider a venue that publishes a fraction of what it receives and forms its sense of merit by reading what it has published. Each decision is defensible on its own terms. No single rejection is wrong. And the venue cannot discover that it erred, because declined work does not return to argue its case and the corpus in which that work appeared does not exist to be compared against.

The structure is common enough to be invisible. It describes journals whose editors were formed by their own back catalogue, curators whose eye was trained by the galleries that selected them, recommender systems fit to engagement logs their own prior recommendations generated, and reward models fit to preferences elicited over samples that earlier reward models scored well. Aggregate homogenization under generative systems is documented from several directions [Padmakumar and He, 2024, Anderson et al., 2024, Doshi and Hauser, 2024], and recursive training on a model's own outputs is now known to erase the tails of the original distribution [Shumailov et al., 2024]. Those results describe an outcome. We ask about the mechanism producing it in a curatorial setting, and about one property of that mechanism which has not been isolated: the filter's training signal is downstream of the filter's own past behavior, and the rejected set, the only evidence that could correct it, is destroyed by construction.

Our setting differs from model collapse in a way that matters for what the result means. There, degradation follows from generated data re-entering training and compounding sampling error. Here, no data is generated by the gatekeeper at all. Every work in the corpus was authored by a creator, and the corpus narrows because of which works were kept. The loop runs through selection rather than generation, which is why the effect survives even though the gatekeeper never writes anything.

The question sits inside this track's framing. The call asks who gains agency through AI-based creative systems and whose agency is absorbed or pre-structured, and asks separately what happens to taste when systems can simulate or predict aesthetic preference. A self-referential gatekeeper is a mechanism by which taste becomes self-confirming with no participant intending it, and the narrowing it produces is authored by nobody in particular.

We contribute an agent-based model isolating the self-referential update as an experimental variable, by comparing an adaptive gatekeeper against a fixed gatekeeper of matched strictness; evidence that the loop narrows the corpus beyond what strictness alone achieves while weakening the statistical signature by which its bias would be audited; and a scope condition, found by sweep after the effect failed to appear under a first specification, establishing that the extra narrowing requires creators who retain positions of their own.

2 Model

2.1 Scope of the operationalization

This is a simulation and the mapping onto any real institution is an argument rather than a measurement. No empirical corpus is analyzed. Creators are points on a sphere rather than people who learn craft, change intent, or leave a field; works are draws around those points rather than objects with internal structure; and one gatekeeper stands in for what is usually a set of overlapping venues. The claim under test concerns the behavior of a mechanism (accept a fraction, retrain on accepted work, discard rejections) under stated assumptions, rather than the sociology of any particular field.

2.2 Creators, works, and quality

A population of $N = 60$ creators occupies the unit sphere in $D = 8$ dimensions, each holding a position c_i initialized uniformly at random. Each round, creator i proposes $w_i = \mathrm{unit}(c_i + \sigma\epsilon)$ with $\sigma = 0.25$ and $\epsilon \sim \mathcal{N}(0, I)$.

Each work carries an intrinsic quality $q_i \sim \mathcal{N}(0, 1)$ drawn independently of position. This is the load-bearing assumption of the paper. Quality and taste are orthogonal by construction, so everything the gatekeeper's filtering does to the corpus is attributable to taste rather than merit. A reader who believes taste should track quality is describing a more forgiving model; we take the harder case, in which the filter has no epistemic warrant for what it excludes.

2.3 The gatekeeper

The gatekeeper holds a taste vector g and accepts the top 25% of each round's proposals by $w_i \cdot g$. Accepted works enter a corpus retaining the most recent 200 items. Rejected works vanish. That disappearance is the mechanism this paper is about: the gatekeeper has no record of what it declined and therefore no signal that could contradict it.

The falsifiable version of our claim is a contrast rather than an effect. Any gatekeeper accepting a quarter of submissions on a consistent criterion will narrow a corpus, so a comparison against the open control alone would establish only that filtering filters. `static` matches strictness exactly and removes the update. Were `adaptive` to match it, the finding would collapse to "selectivity costs breadth," a claim about strictness rather than self-reference, which is why the design has four arms and not two.

Table 1: The four conditions. All share acceptance fraction, corpus window, creator dynamics, and seeds, differing only in how the gatekeeper selects and updates. `static` is the load-bearing control, matching strictness while removing the self-referential update.

Condition	Gatekeeper behavior
`open`	Accepts a uniformly random 25%; no taste at all. Establishes the breadth ceiling.
`static`	Fixed taste $g = g_0$, never updated. Isolates strictness from the loop.
`adaptive`	Re-estimates g each round as the mean of its own accepted corpus.
`adaptive_quota`	Adaptive, with 20% of acceptance slots filled uniformly at random from the rejected pool.

2.4 Creators adapt, and also resist

Creators observe the accepted set and drift toward its centroid at rate $\mu = 0.06$, while being pulled back toward their own initial position at rate $\alpha = 0.10$:

$$c_i \leftarrow \mathrm{unit}(c_i + \mu\,\mathrm{unit}(\bar{a}) + \alpha c_{i,0}) ,$$

where $\bar{a}$ is the mean of the round's accepted works. The mimicry term is a second channel of narrowing and it is voluntary, which matters for the agency argument: creators are not filtered into conformity so much as they walk toward it, in the manner of an informational cascade where each individual inference from others' revealed choices is locally rational [Bikhchandani et al., 1998]. The anchor term is what makes a creator someone with a standing commitment rather than a pure imitator.

2.5 Anchor calibration (disclosed)

The anchor was added in response to a result, and its provenance belongs in the method. Our first specification had none. Under it the creator population collapsed to a single point by round 50 (spread < 0.003), every condition returned an identical corpus breadth near 0.141, and the `adaptive`–`static` contrast was null ($p = 0.61$). Diagnosis showed collapse rather than absence of effect: with all creators at one point there is nothing for any filter to select among. The gatekeeper's taste was in fact moving throughout, a mean cosine distance of 0.24 from its origin over a run and up to 0.40, so the loop was running and had nothing to bite on. The anchor prevents collapse. Section 3.5 sweeps it jointly with μ and reports where each finding holds; that sweep was run after the reported main results and was not used to select them.

2.6 Measures

Corpus breadth. Mean pairwise cosine distance among works in the gatekeeper's corpus. This is what narrowing means operationally, and it does not depend on attributing works to creators.

Creator spread. Mean pairwise cosine distance across the creator population, measuring whether the field itself, rather than only its published record, has converged.

Survivor bias. Pearson correlation across creators between $1 - c_{i,0} \cdot g_0$, distance from the gatekeeper's founding taste, and realized acceptance rate over the run. A strongly negative value means distance from the founding position reliably predicted exclusion, which is the signature an outside analyst would look for when auditing a filter for bias.

Confidence intervals are bootstrap percentile intervals over runs, with runs as the independent unit; p-values are two-sided permutation tests on the difference in means [Efron and Tibshirani, 1993].

3 Results

3.1 The loop narrows beyond strictness (headline)

The self-referential gatekeeper produces a narrower corpus than the fixed gatekeeper of identical strictness (Table 2, Figure 1A): 0.423 (95% CI 0.417–0.429) against 0.496 (0.487–0.506), difference -0.074, $p = 0.0001$, both $n = 40$. Both sit far below the open ceiling of 0.804 ($p = 0.0001$ for

Table 2: Main results, $n = 40$ runs per condition, 200 rounds each; brackets are bootstrap 95% CIs. Higher corpus breadth is more diverse. Survivor bias closer to zero means exclusions are less predictable from a creator's founding position. The adaptive gatekeeper has the narrowest corpus of all four and less than half the measurable bias of the static one.

Condition	Corpus breadth	Creator spread	Survivor bias r
`open`	0.804 [0.798, 0.811]	0.728 [0.719, 0.735]	0.029 [−0.009, 0.070]
`static`	0.496 [0.487, 0.506]	0.764 [0.755, 0.773]	−0.937 [−0.941, −0.933]
`adaptive`	**0.423** [0.417, 0.429]	0.724 [0.715, 0.733]	−0.497 [−0.559, −0.431]
`adaptive_quota`	0.523 [0.515, 0.530]	0.726 [0.716, 0.735]	−0.442 [−0.516, −0.361]

`adaptive` versus `open`). The control is load-bearing here: matched strictness with a fixed taste does not reach the same narrowness, so the extra 0.074 is attributable to the update rather than to selectivity.

The mechanism is visible in the gatekeeper's trajectory. Under `adaptive`, g drifts a mean cosine distance of 0.24 from its starting position over a run. It moves because it follows its corpus, and the corpus is what it selected. Nothing external pulls it back.

3.2 Illegibility: the narrower filter is the one that audits clean

We expected the adaptive gatekeeper to look worse on every axis. It does not (Figure 1C). Survivor bias under `static` is -0.937 (CI -0.941 to -0.933): distance from the founding taste predicts exclusion almost perfectly. Under `adaptive` the same correlation is -0.497 (CI -0.559 to -0.431), roughly half the magnitude, intervals non-overlapping.

By the measure an auditor would reach for, the adaptive gatekeeper is the fairer one, and it is also the one publishing the narrowest corpus. The reason is that its exclusion zone moves. A fixed gatekeeper excludes the same region for two hundred rounds, and that stability is exactly what a correlation against founding position detects. An adaptive gatekeeper's excluded region drifts with its taste, so no single fixed reference predicts who was left out, and the bias disperses across a trajectory rather than sitting still to be measured. Legibility of bias and breadth of outcome are separate quantities, and here they point in opposite directions.

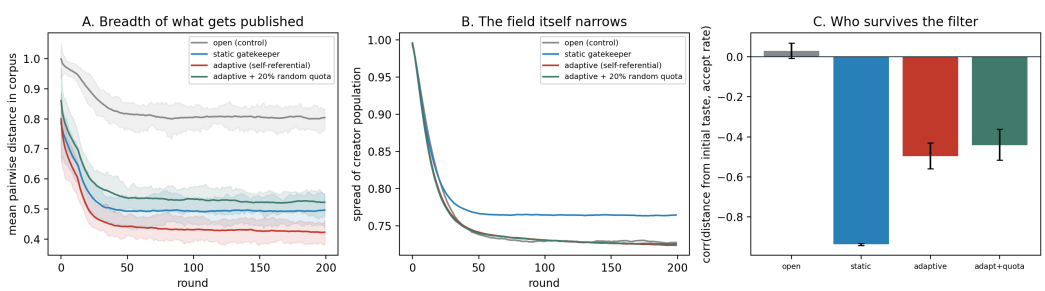

Figure 1: Corpus breadth over 200 rounds (A), creator spread (B), and survivor bias by condition (C); $n = 40$ runs per condition, bootstrap 95% CIs, shaded bands in A are percentile intervals across runs. The adaptive gatekeeper (red) settles to the narrowest corpus in A while showing roughly half the measurable bias of the static gatekeeper (blue) in C. Panel B is the null of §3.4: creator spread is flat across gatekeeper conditions, so the narrowing in this regime happens in the published record rather than in the population.

3.3 Quotas buy breadth, not openness

Reserving 20% of acceptances for uniformly random selection raises corpus breadth from 0.423 to 0.523 ($+0.100$, $p = 0.0001$), recovering about a quarter of the distance to the open ceiling. Survivor bias moves only from -0.497 to -0.442, with overlapping intervals.

The quota does what it is built to do and nothing more. It injects variety into the published record without changing who the gatekeeper is or where its taste is travelling. Its effect is on the corpus, not on the loop.

3.4 A null: the field itself does not converge

Creator spread is flat across gatekeeper conditions (Figure 1B): 0.728, 0.764, 0.724, and 0.726 for `open`, `static`, `adaptive`, and `adaptive_quota`. The `static` condition is nominally highest, the opposite of what a story about gatekeepers homogenizing creators would predict.

We report this as a null and read it as a property of the parameter regime rather than a claim about the world. With $\alpha = 0.10$ against $\mu = 0.06$ the anchor dominates mimicry, so creators largely hold their ground whatever gets published, and narrowing in this regime happens in the record rather than in the population. Section 3.5 shows what the opposite regime looks like.

3.5 Sensitivity, and where the effect lives

We swept $\alpha \in \{0, 0.05, 0.10, 0.20\}$ against $\mu \in \{0.02, 0.06, 0.12\}$ at $n = 20$ runs per cell, tracking two findings: F1, that `adaptive` narrows further than `static`; and F2, that `adaptive` shows weaker survivor bias than `static`.

F2 holds in 12 of 12 cells, with the static correlation between -0.508 and -0.954 and the adaptive one between -0.395 and -0.671 throughout. The illegibility result is not an artifact of parameter choice.

F1 holds in 10 of 12. Both failures occur at $\alpha = 0$, precisely where creator spread collapses to 0.002 or below. This is the regime our first specification was trapped in, so the sweep confirms the diagnosis rather than merely repeating it. We state the consequence as a scope condition: the loop's extra narrowing requires a field whose members retain positions of their own. In a population of pure imitators, homogenization has already occurred by other means and the gatekeeper's update is irrelevant to an outcome already determined.

4 Discussion

The two results pull against each other, and that tension is the point. An institution that updates its taste from its own outputs publishes a narrower body of work than one holding a fixed line, and it looks more even-handed while doing so. If the available audit is a correlation between some fixed notion of position and acceptance, the drifting filter passes and the static one fails, exactly inverting their effect on the corpus. Any fairness instrument anchored to a fixed reference point will be least sensitive to precisely the filters whose criteria move.

This bears on the call's question about accountability when agency is distributed across artists, models, datasets, tools, and institutions. In this model nobody authors the narrowing. The gatekeeper follows its corpus, the corpus records its decisions, and creators drift toward what gets published, voluntarily. Every local step is defensible and the aggregate is a field that has lost most of its range, with no decision available to point at as the error. That is a different shape of problem from bias in the ordinary sense, where there is at least a criterion one could name and contest.

Practically, the intervention we test recovers breadth and leaves the dynamic intact. What would address the dynamic is a record of rejections that feeds back into the filter, which is the one thing this structure destroys. Institutions that do keep such records, through published dissents, reviewer disagreement data, or preserved rejection archives, are maintaining exactly the counterfactual this model deletes, and the shape of that maintenance is familiar from durable commons institutions [Ostrom, 1990].

One empirical analogue is worth naming. Salganik et al. [2006] built an artificial music market and found that increasing social influence raised both inequality and unpredictability of success, with quality only partly determining outcomes. Their mechanism is peer visibility rather than institutional gatekeeping, and their unpredictability is across worlds rather than within one. The convergence is that both results locate a substantial part of what a field ends up valuing in the structure of the process rather than in the merit of the work, and our survivor bias measure is a within-run analogue of the question their multiple-worlds design answers across runs.

5 Limitations

Simulation. No empirical corpus was analyzed, and the mapping to real institutions is argument rather than measurement. **Orthogonal quality.** Quality is constructed independent of gatekeeper taste so that filtering cannot be justified on merit; if taste tracks quality even partially, the narrowing carries information and the normative reading changes substantially. **Anchor provenance.** The anchor term was introduced after a null result, a researcher degree of freedom exercised in response to data; it is disclosed in §2.5 and swept in §3.5, which bounds the concern without eliminating it. **Unswept structure.** Acceptance fraction, corpus window, population size, dimensionality, and proposal noise were held fixed throughout; only α and μ were varied. **Audit specificity.** Survivor bias is measured against founding taste, so the illegibility claim is specific to a fixed-reference audit; an analyst tracking the gatekeeper's current taste each round would detect more. **Single gatekeeper.** Real fields have overlapping venues with differing tastes, and a rejected work can go elsewhere, which is the most consequential simplification here.

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